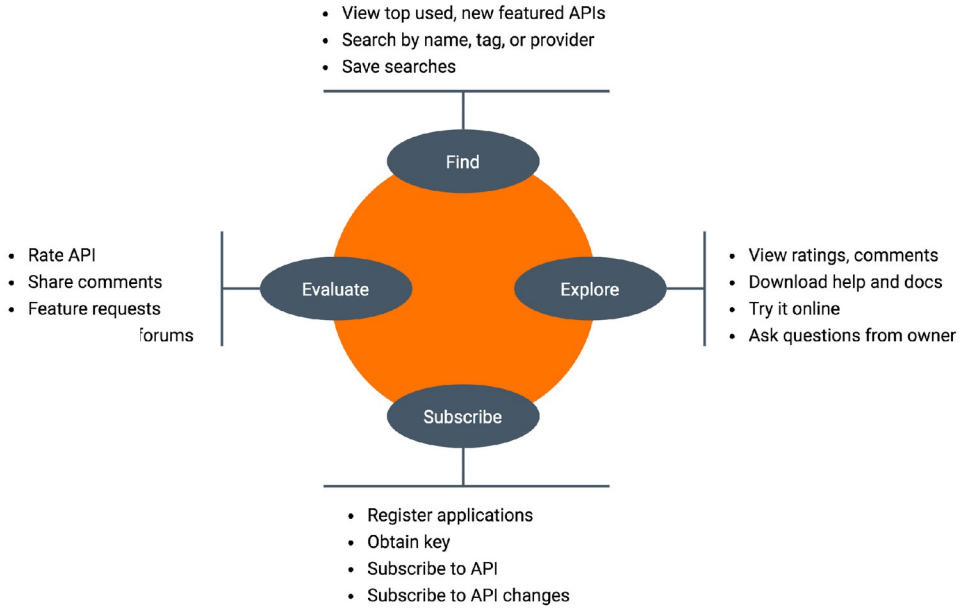
Developer Portal Visibility

WSO2

What is API Developer Portal?

- React Web Application
- API Publishers can
  - Host APIs
  - Advertise APIs
- API Consumers can
  - Self Register
  - Discover APIs
  - Evaluate APIs
  - Subscribe to APIs
  - Consume APIs



Discover APIs

- Anonymous (Public) View
- Logged in View
- Requirements for an API to be displayed in the Developer Portal

  API should be in Published state

  API should be indexed in solr

  API should be either a public API or the logged in user should have permission to view the API
- Search APIs by

  API Name, Provider, Version, Context, Status, Description, Tags, API Category and Custom Property. (prefix based search e.g., name:Pizza)

  Unified content search (default)

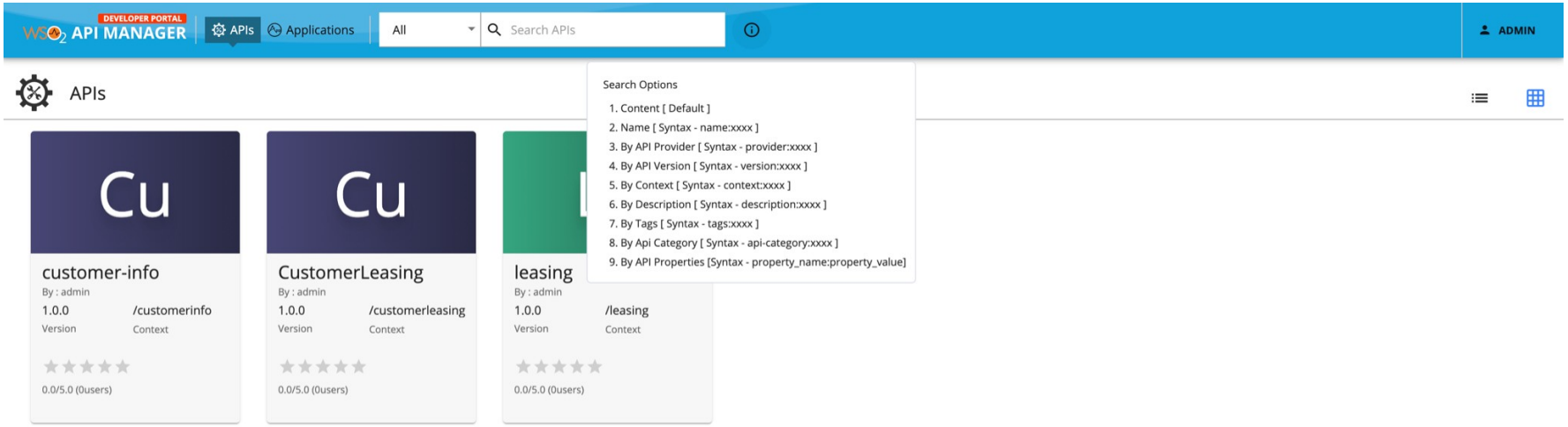
The API Search Facility

Search for an API by,

- API name
- API version
- Description
- API provider
- Context
- API Category
- API status
- Tag
- Custom property

Link - API Search

Discover APIs



Developer Portal Visibility

WSO2

Developer Portal Visibility

- Control API visibility in the Developer Portal
- Control Subscription availability in Developer Portal
- Control Gateway visibility in Developer Portal
- Organization-aware Developer Portal (B2B API Consumption)

Gateway Visibility in Developer Portal

- API developers need to control access to the gateway environments deployed in their APIs for the application developers.
- Access to Gateways can be restricted based on user roles either from the Admin Portal or via the <API-M_HOME>/repository/conf/deployment.toml file.
- When the permission type for the role-based restriction is set to **ALLOW**, only users with the specified roles are permitted to use the gateway environment at the Developer Portal.
- Similarly, when the permission type for the role-based restriction is set to **DENY**, users with the specified roles are not allowed to use the gateway environment at the Developer Portal.

Link - Configure Gateway Visibility

B2B API Consumption

- In traditional setups without organization-aware Identity Providers (IDPs), the Developer Portal operates with a flat user hierarchy.

- This limits enterprise use cases by lacking:
  - API visibility control across organizational boundaries
  - Organization-specific subscription and business plans

- With B2B API Consumption in WSO2 API Manager:
  - When the IDP provides organization claims, the Developer Portal becomes organization-aware
  - API publishers can:
    - Restrict API visibility to selected organizations
    - Define custom business plans per organization

Link - B2B API Consumption

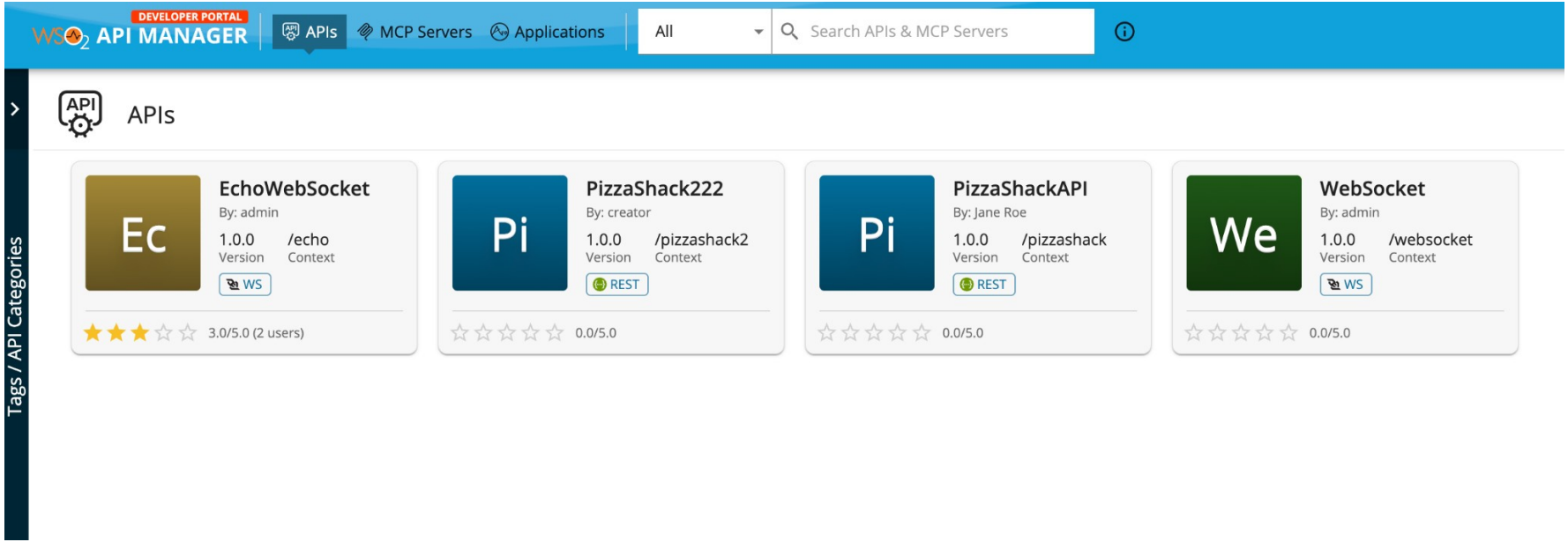
How B2B API Consumption Works?

- IDPs are configured to include organization claims during authentication (Link)
- Organizations and their hierarchy are registered in WSO2 API Manager (Link)
- API Product Managers can:
  - Publish APIs to selected organizations (Link)
  - Offer differentiated business plans
- API Consumers log in to the Developer Portal can:
  - See APIs shared to their organization
  - Subscribe to APIs using available business plans for their organization
  - Share applications between organizations

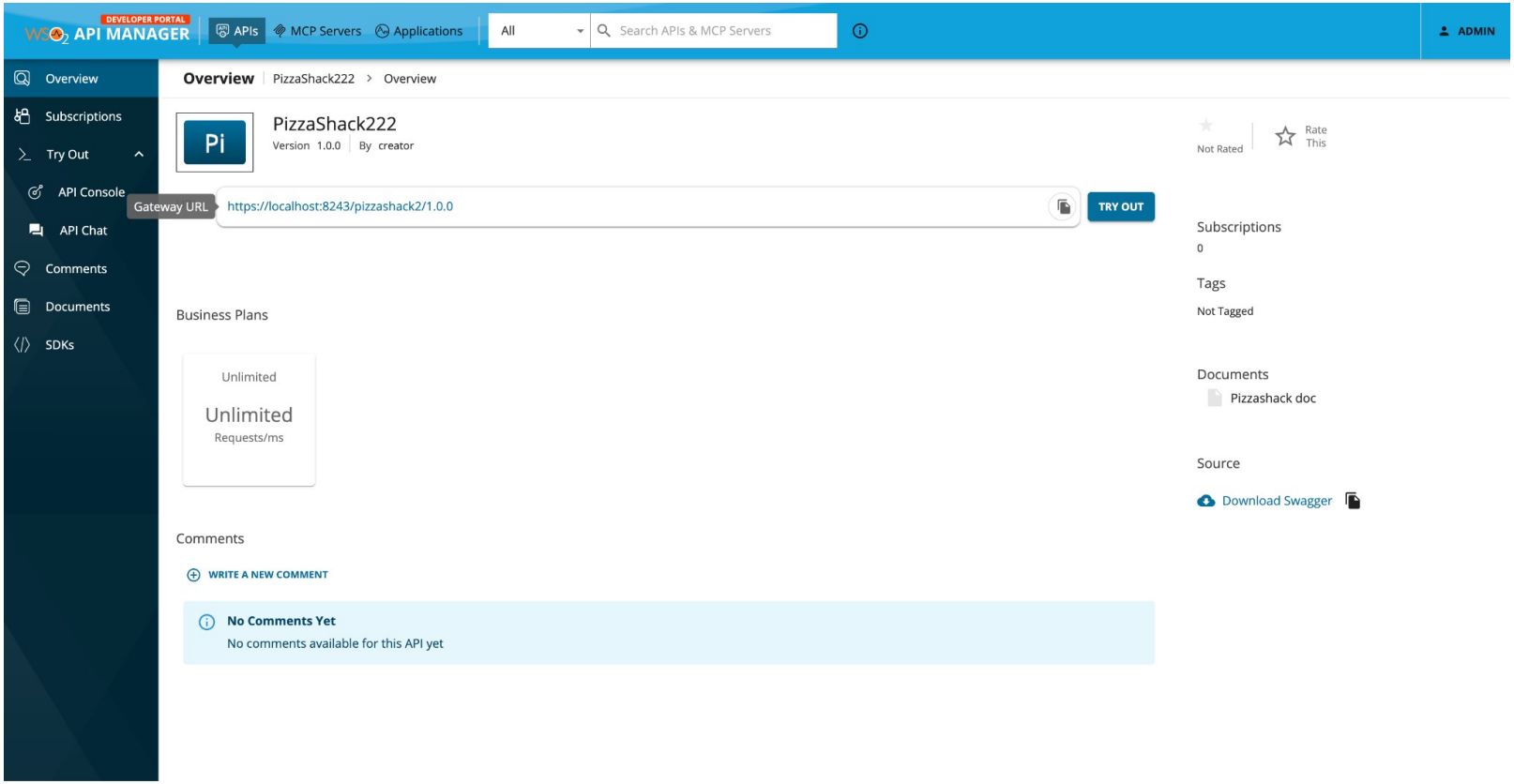
APIs and Applications

WSO2

API Listing


DEVELOPER PORTAL
API MANAGER
APIs
MCP Servers
Applications
All
Search APIs & MCP Servers
APIs
Tags / API Categories
Ec
EchoWebSocket
By: admin
1.0.0
Version
/echo
Context
WS
3.0/5.0 (2 users)
Pi
PizzaShack222
By: creator
1.0.0
Version
/pizzashack2
Context
REST
0.0/5.0
Pi
PizzaShackAPI
By: Jane Roe
1.0.0
Version
/pizzashack
Context
REST
0.0/5.0
We
WebSocket
By: admin
1.0.0
Version
/websocket
Context
WS
0.0/5.0

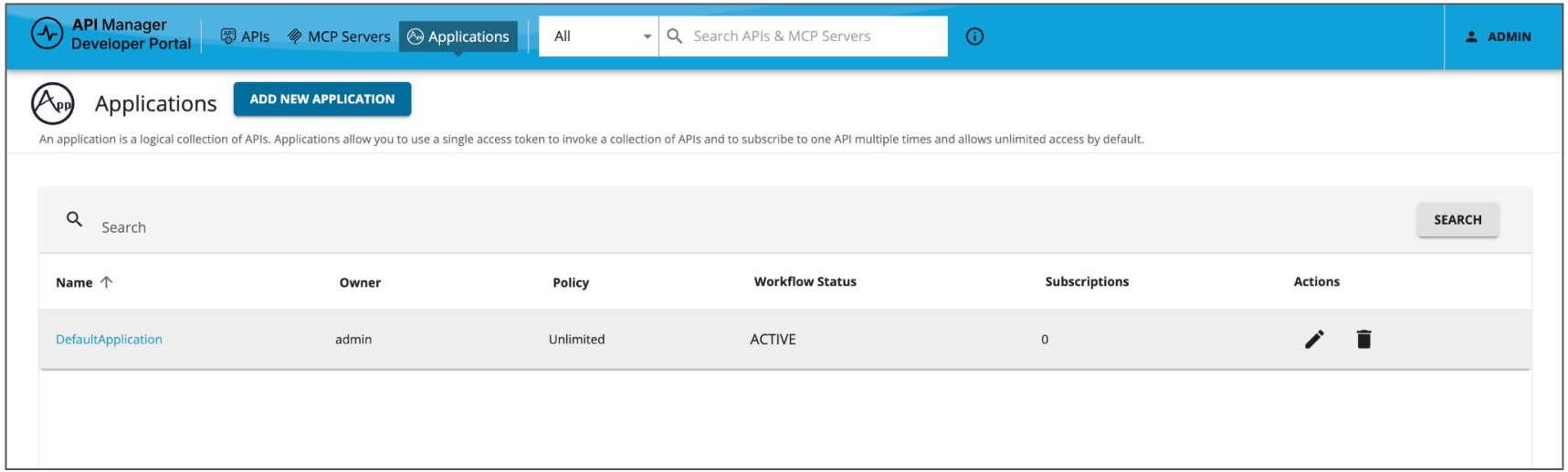
API Overview



Applications

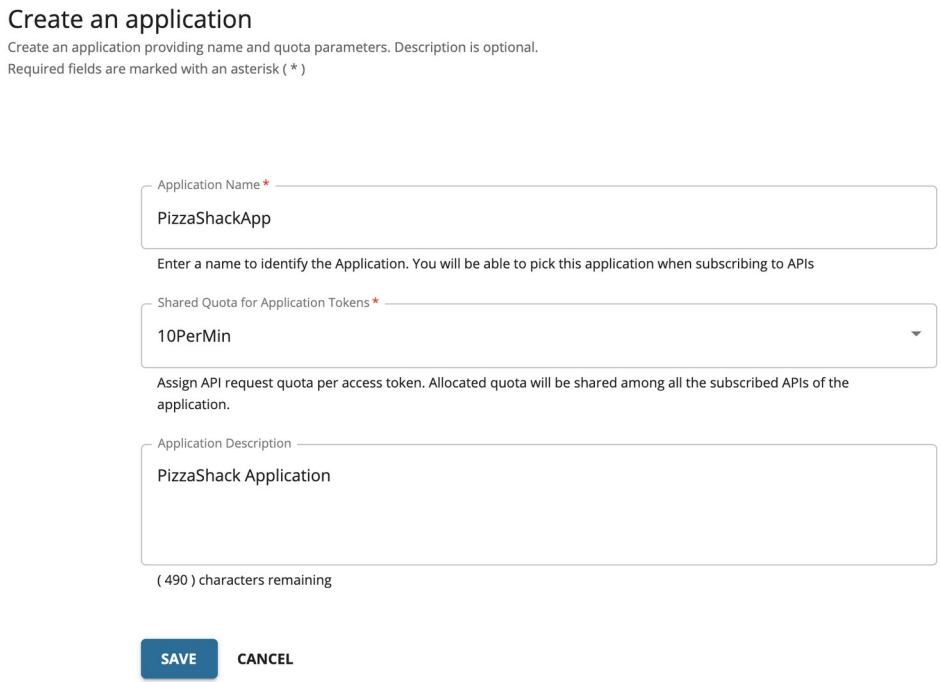
- Logical representation of an application such as a mobile app, web app, device, etc.
- Generate and use a single key for multiple APIs.
- Subscribe multiple times to a single API with different Service Level Agreements (SLAs)/ business plans which operate on per access token basis
- Comes with a pre-created default application, which allows unlimited access by default.

Application Creation



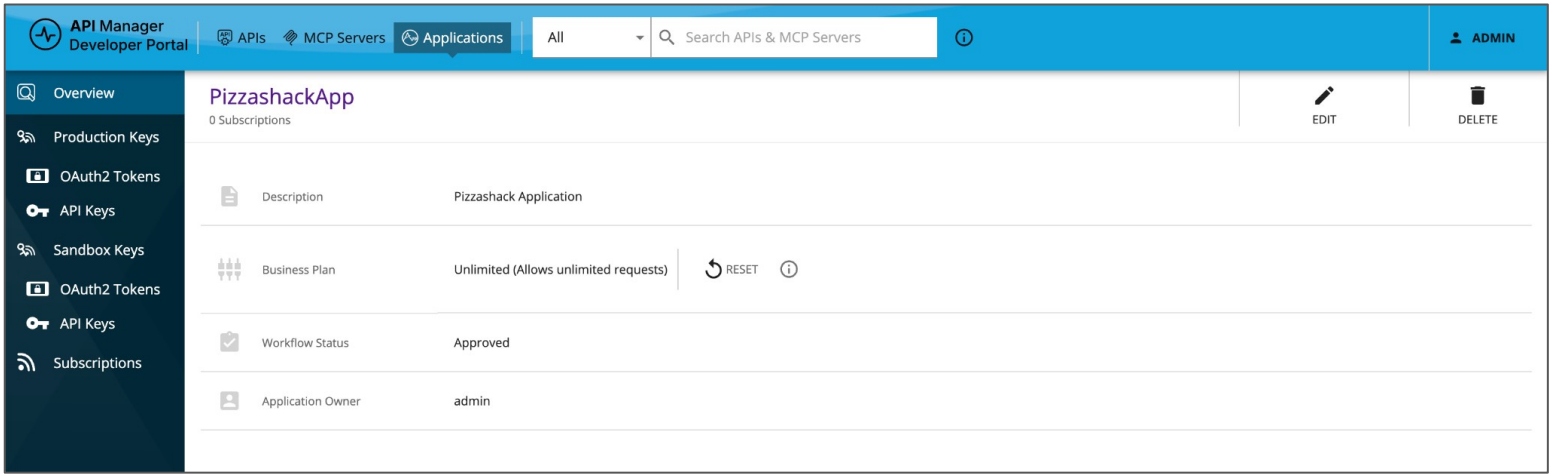
Application Creation

- Create Application Form



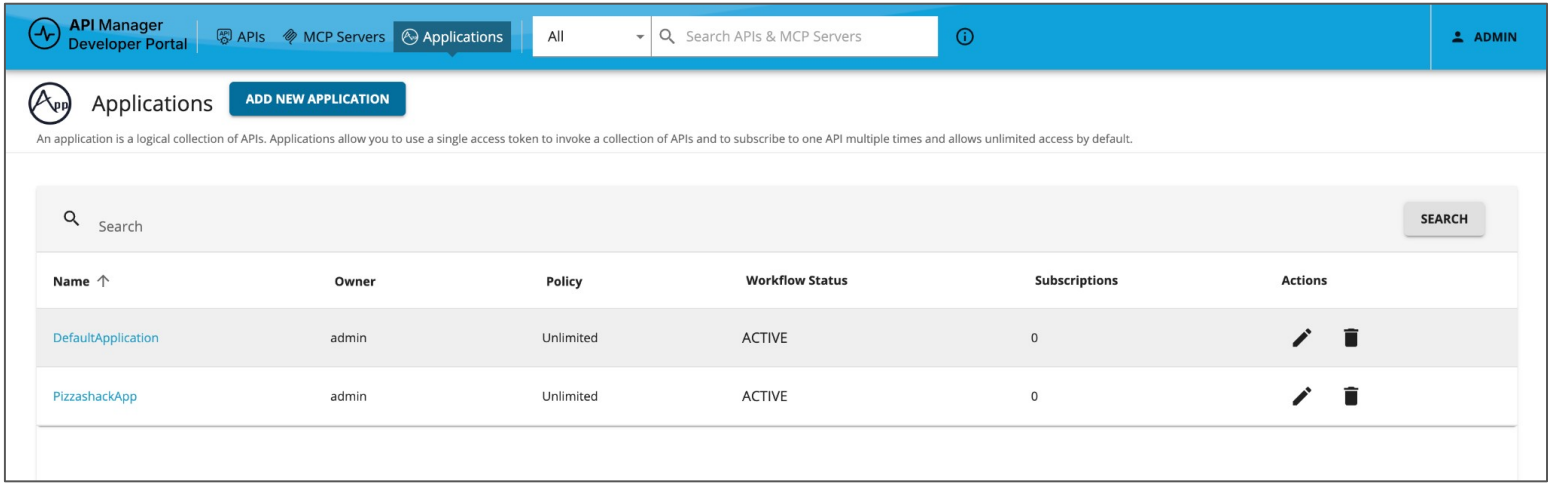
Application Overview

- Application Overview



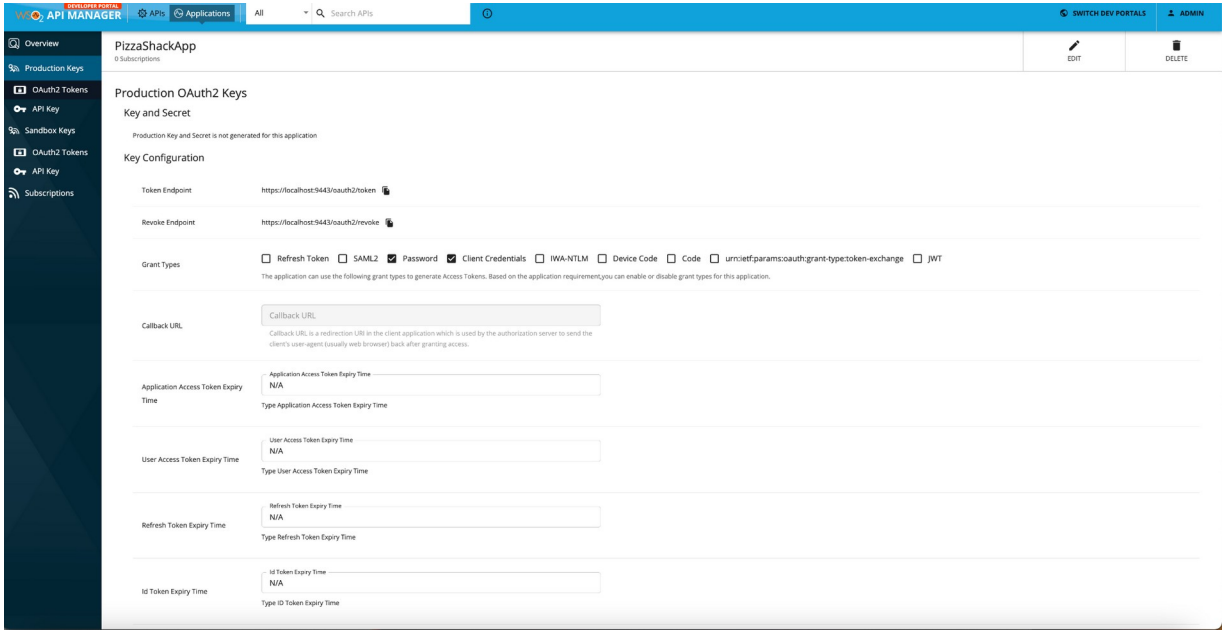
Application Listing

- Application Listing

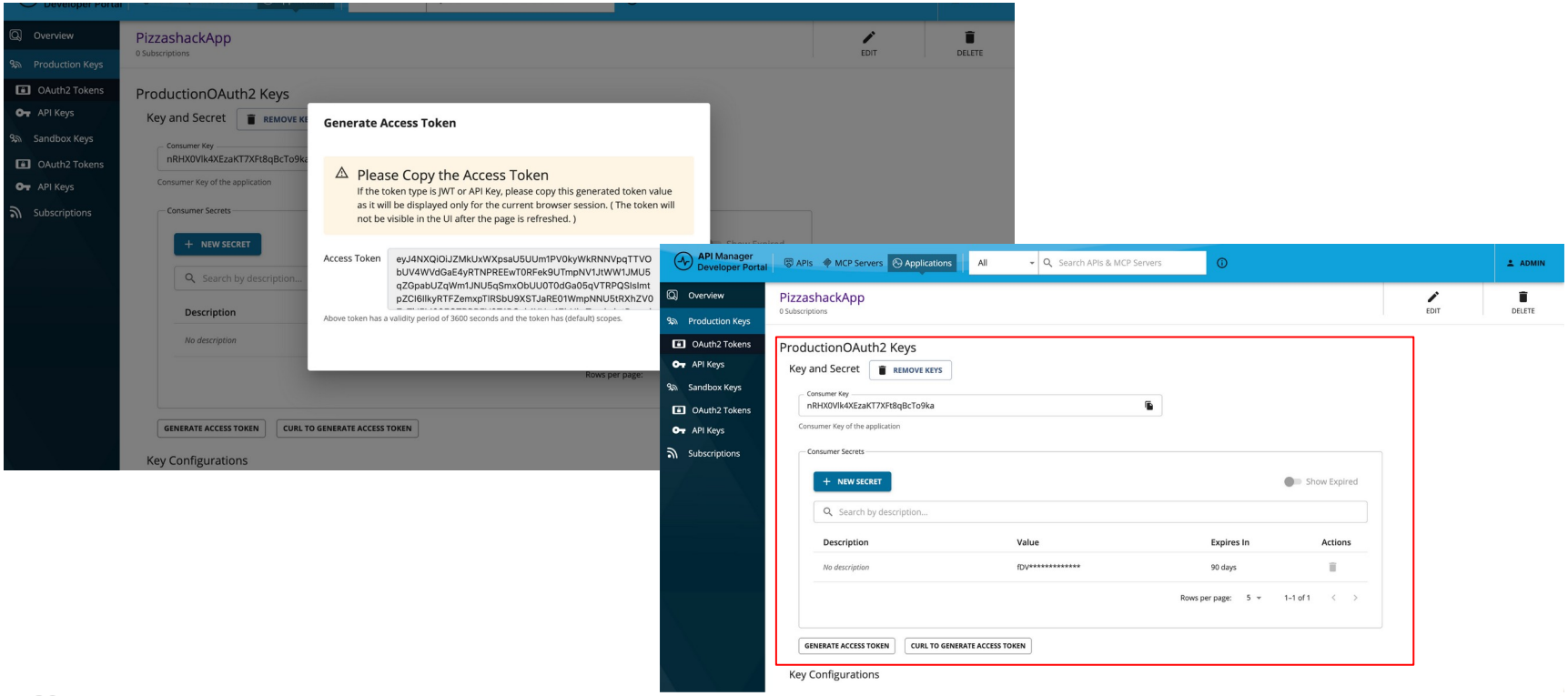


Generate Keys

- Can be Production or Sandbox keys.
- Generates **consumer-key and consumer-secret** pair for the application.



Generate Keys



Supported Grant Types

- Supported Grant types:
  - Password Grant
  - Client Credentials Grant
  - Authorization Code Grant
  - Refresh Token Grant
  - JWT Grant
  - SAML Extension Grant
  - Kerberos OAuth2 Grant
  - NTLM Grant

Supported Grant Types

Password (legacy): Used by first-party clients to exchange a user's credentials for an access token. Since this involves the client asking the user for their password, it should not be used by third party clients. In this flow, the user's username and password are exchanged directly for an access token.

```
POST /oauth2/token HTTP/1.1
Host: authorization-server.com

grant_type=password
&username=user@example.com
&password=1234luggage
&client_id=xxxxxxxxxx
&client_secret=xxxxxxxxxx
```



```
HTTP/1.1 200 OK
Content-Type: application/json
Cache-Control: no-store
Pragma: no-cache

{
  "access_token":"MTQ0NjJkZmQ5OTM2NDE1ZTZjNGZmZjI3",
  "token_type":"bearer",
  "expires_in":3600,
  "refresh_token":"IwOGYzYTlmM2YxOTQ5MGE3YmNmMDFkNTVk",
  "scope":"create"
}
```

Supported Grant Types

Client Credentials: Used by clients to obtain an access token outside of the context of a user. This is typically used by clients to access resources about themselves rather than to access a user's resources.

```
POST oauth2/token HTTP/1.1
Host: authorization-server.com

grant_type=client_credentials
&client_id=xxxxxxxxxx
&client_secret=xxxxxxxxxx
```



```
HTTP/1.1 200 OK
Content-Type: application/json
Cache-Control: no-store
Pragma: no-cache

{
  "access_token":"MTQ0NjJkZmQ5OTM2NDE1ZTZjNGZmZjI3",
  "token_type":"bearer",
  "expires_in":3600,
  "refresh_token":"IwOGYzYTlmM2YxOTQ5MGE3YmNmMDFkNTVk",
  "scope":"create"
}
```

Supported Grant Types

Authorization Code: Used by confidential and public clients to exchange an authorization code for an access token. After the user returns to the client via the redirect URL, the application will get the authorization code from the URL and use it to request an access token.

```
POST /oauth2/token HTTP/1.1
Host: authorization-server.com

grant_type=authorization_code
&code=xxxxxxxxxx
&redirect_uri=https://example-app
.com/redirect
&client_id=xxxxxxxxxx
&client_secret=xxxxxxxxxx
```



```
HTTP/1.1 200 OK
Content-Type: application/json
Cache-Control: no-store
Pragma: no-cache

{
  "access_token":"MTQ0NjJkZmQ5OTM2NDE1ZTZjNGZmZjI3",
  "token_type":"bearer",
  "expires_in":3600,
  "refresh_token":"IwOGYzYTlmM2YxOTQ5MGE3YmNmMDFkNTVk",
  "scope":"create"
}
```


Demo: https://www.oauth.com/playground/authorization-code.html

Supported Grant Types

Refresh Token: Used by clients to exchange a refresh token for an access token when the access token has expired. This allows clients to continue to have a valid access token without further interaction with the user.

```
POST /oauth2/token HTTP/1.1
Host: authorization-server.com

grant_type=refresh_token
&refresh_token=xxxxxxxxxxx
&client_id=xxxxxxxxxx
&client_secret=xxxxxxxxxx
```



```
HTTP/1.1 200 OK
Content-Type: application/json
Cache-Control: no-store
Pragma: no-cache

{
  "access_token":"MTQ0NjJkZmQ5OTM2NDE1ZTZjNGZmZjI3",
  "token_type":"bearer",
  "expires_in":3600,
  "refresh_token":"IwOGYzYTlmM2YxOTQ5MGE3YmNmMDFkNTVk",
  "scope":"create"
}
```

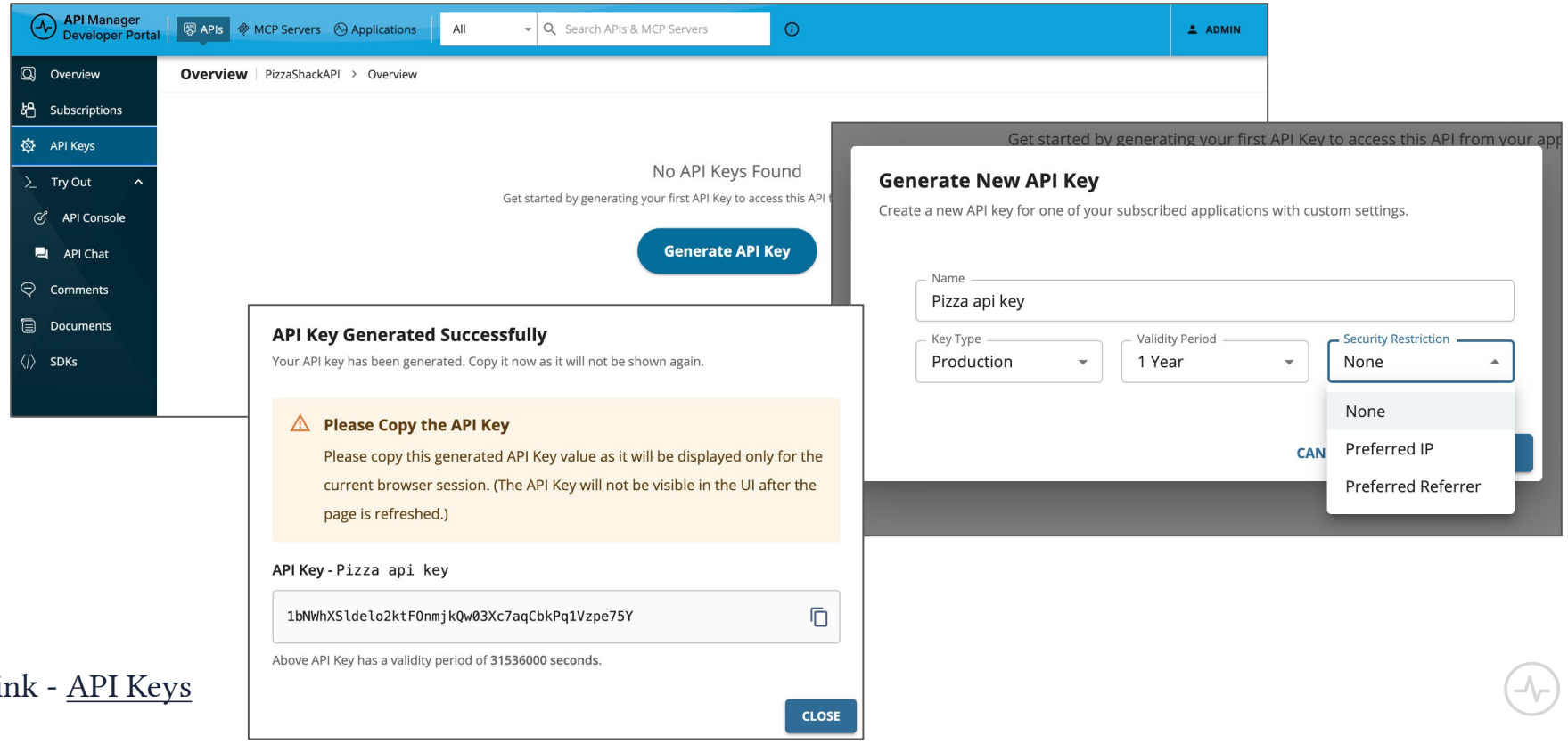
Access Tokens

- Application Access Tokens : Tokens to identify and authenticate an entire application
- User Access Tokens : Tokens to identify the final user of an application.

JWT Token



API Key



Link - API Keys

Subscribing and Invoking the API

WSO2

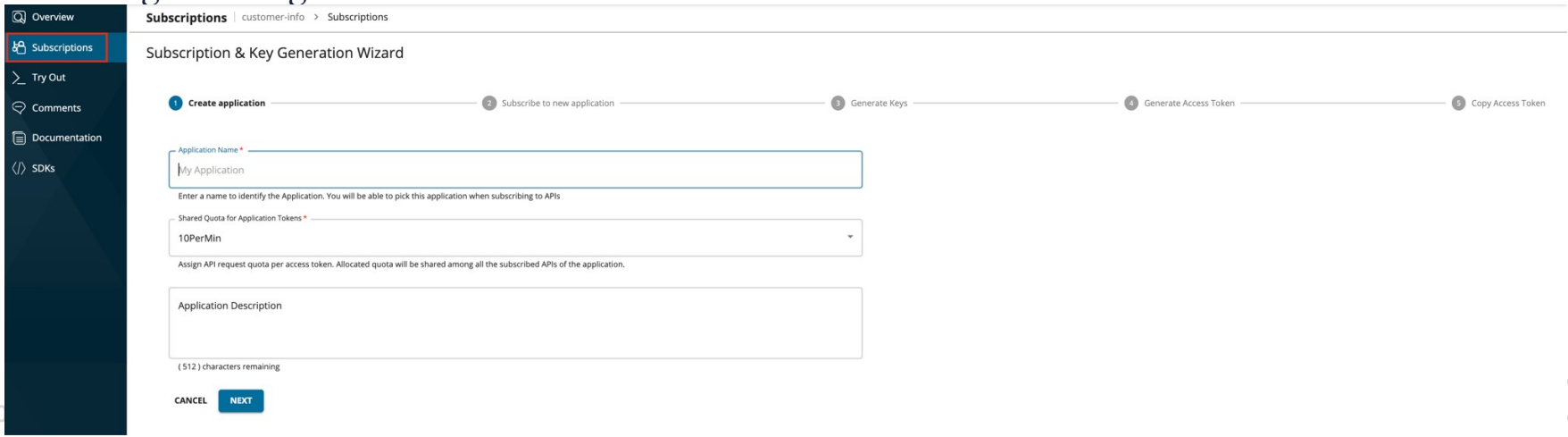
Subscribing an API

There are three (03) ways an API can be subscribed.

1. Using the **API Subscription and Key Generation Wizard**.
    a. A new application will be created and subscribed
2. Subscribing to an existing application.
    a. Select an existing application to subscribe
3. Using **Try Out** in overview page.
    a. An API subscription will be added for the Default Application.

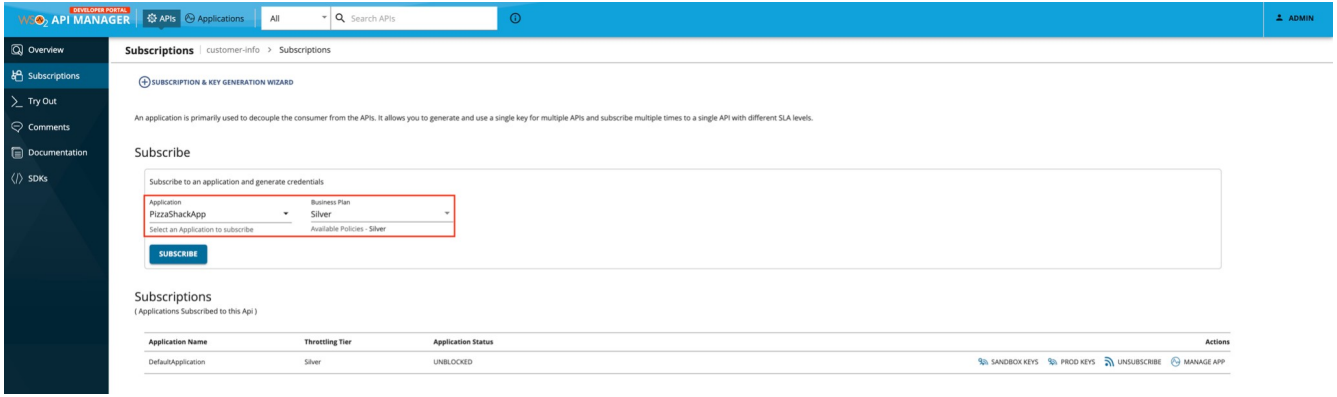
Subscription and Key Generation Wizard.

- Navigate to **Subscriptions** tab.
- Click on the **Subscription and Key Generation Wizard**.
- This will guide you with creating a new application, subscribing to the API and generating tokens.

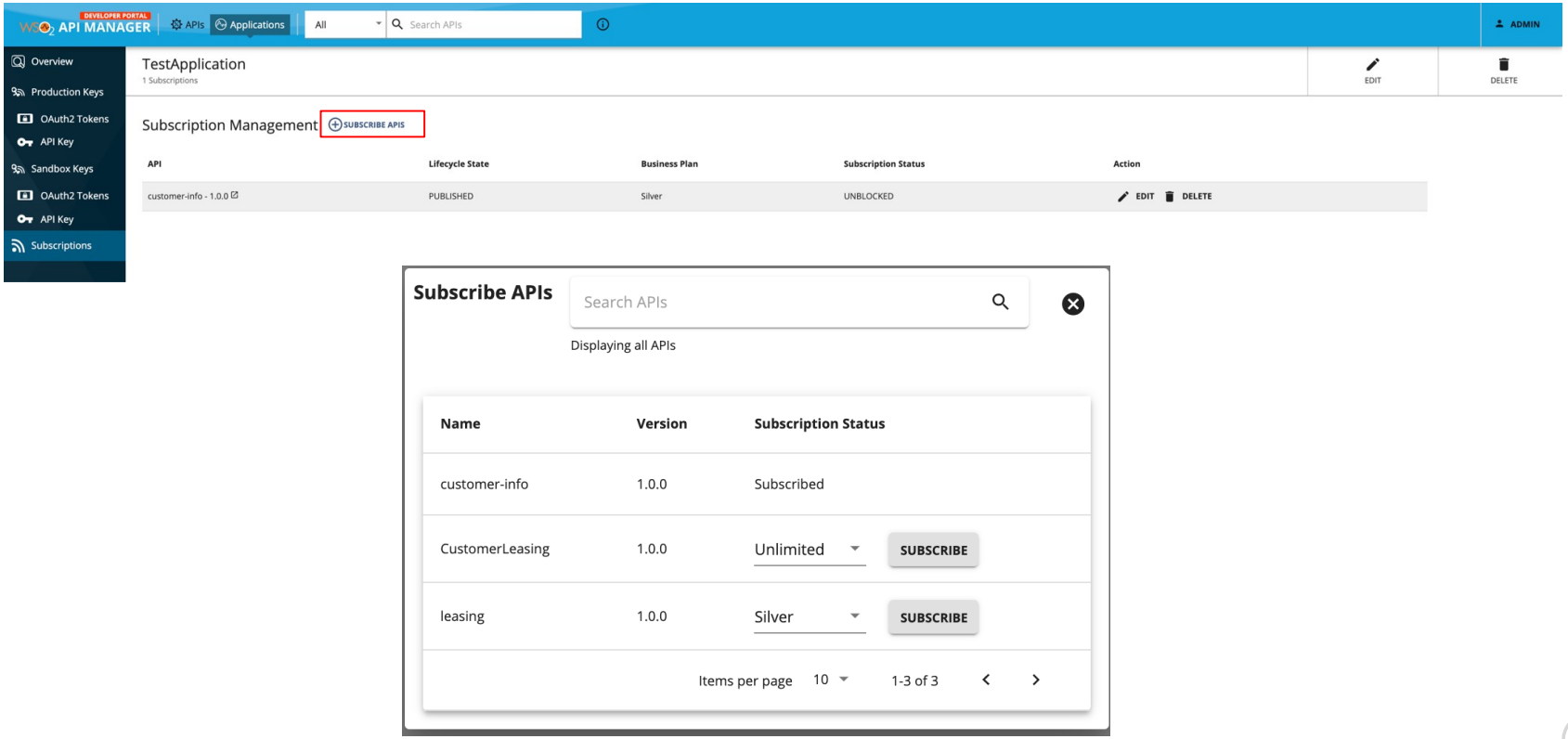


Subscribing to an Existing Application

- Open an API, and go to the **Subscriptions** page by clicking the **Subscriptions** item in the left menu.
- Under the **Subscribe** section, select the application which is created before and click **Subscribe**.

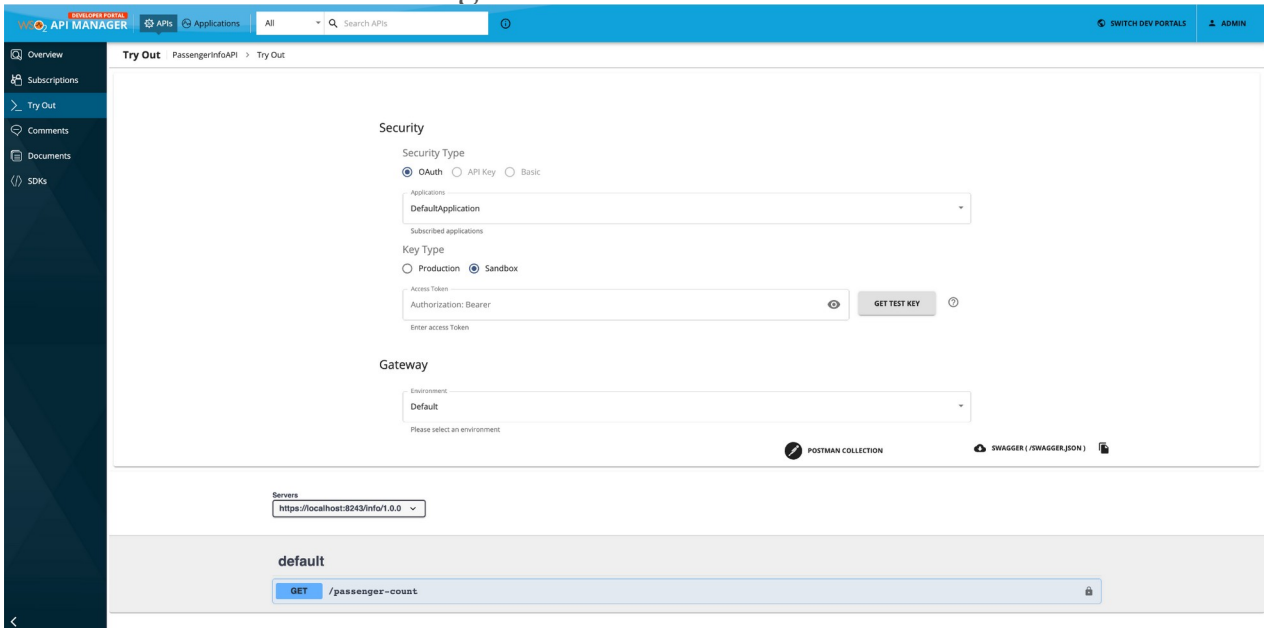


Add Subscription to an API in the Application

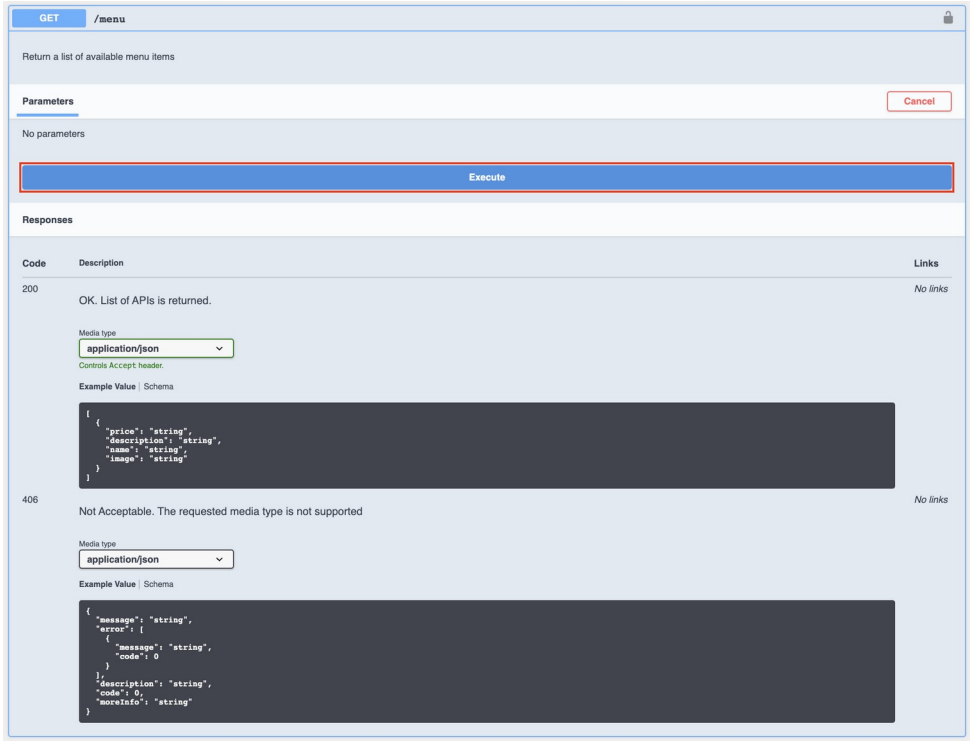


Invoke API

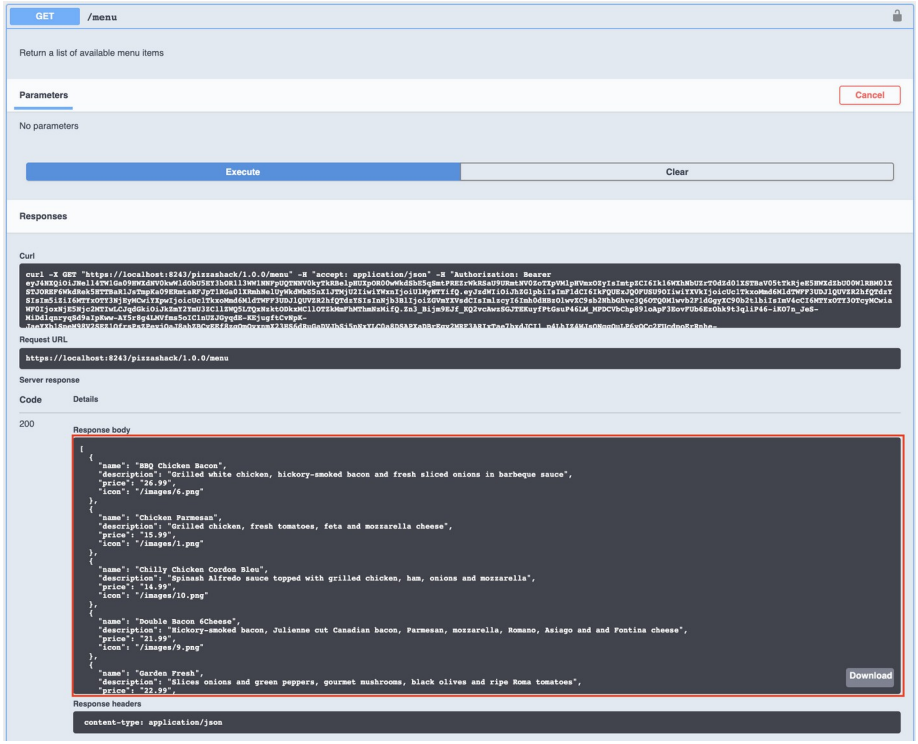
- API should be subscribed to an Application.
- Access token needs to be generated to access the API resources.



Invoke an API



Invoke an API



Application Sharing

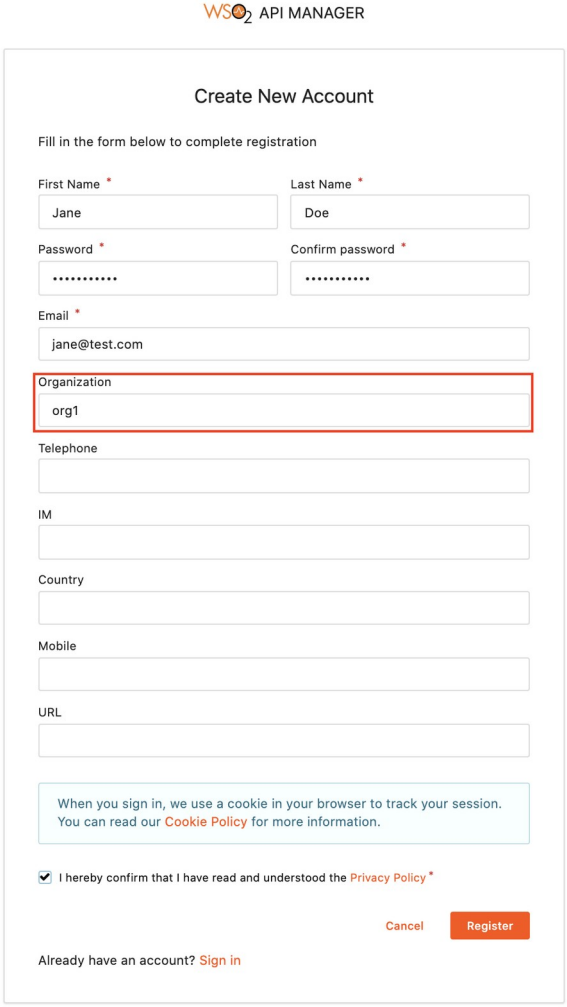
WSO2

Application Sharing

- Enable application sharing in configure deployment.toml to use the feature.

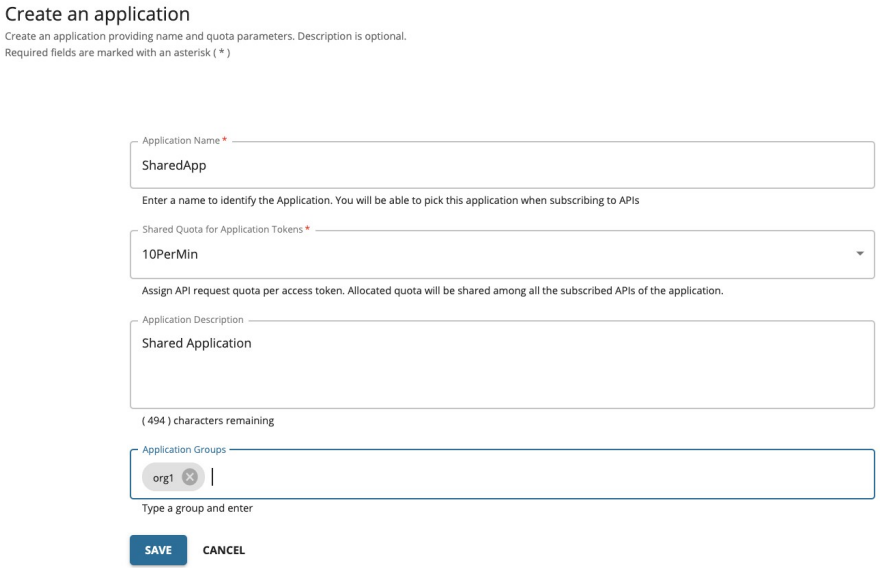
```
[apim.devportal]
enable_application_sharing = true
```

- Register users with Organizations



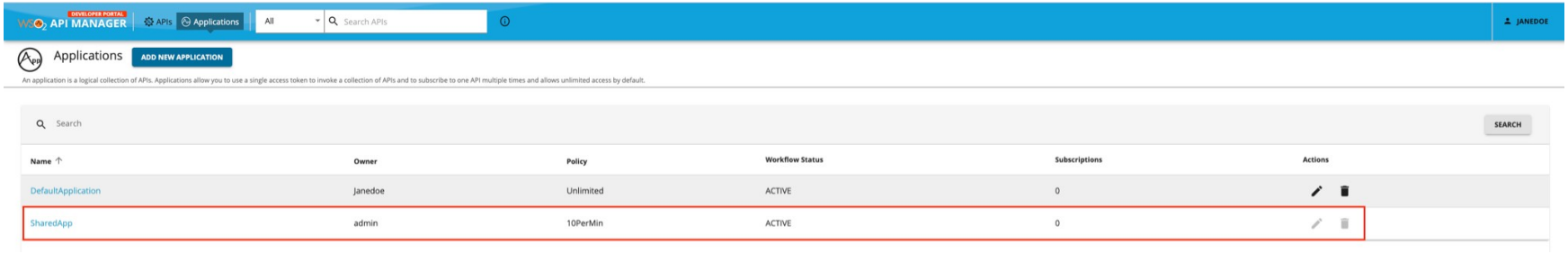
Application Sharing

- Create an Application with Application Groups defined.



Application Sharing

- Users who belong to that specific organization can use that Application.
- Only owner of that application can generate Application keys.



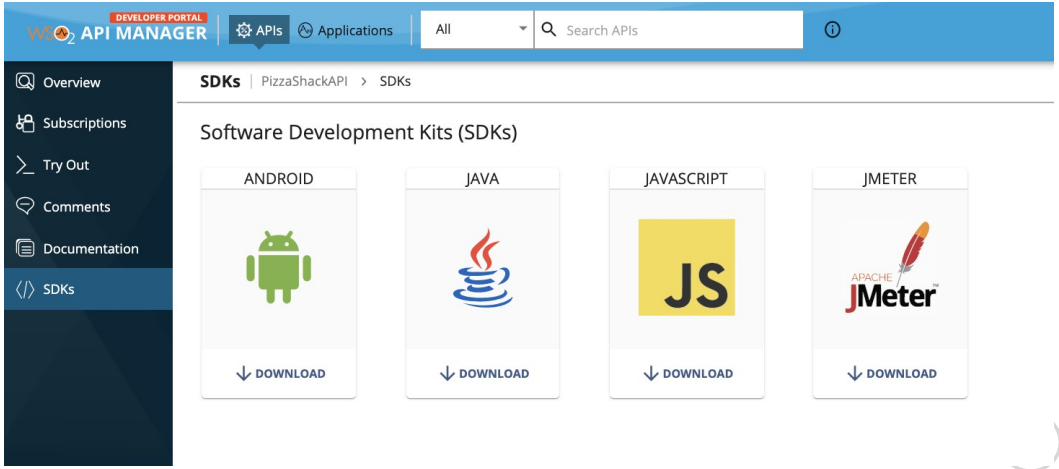
Client SDKs

WSO2

Client SDKs

- With SDKs, developers can implement API Clients which can be used to invoke the particular API from different platforms (eg: android, web etc).
- The supported SDK platforms can be configured in the deployment.toml file.

```
[apim.sdk]
supported_languages = ["android", "java", "scala",
"csharp", "dart", "flash", "groovy", "javascript"]
```



 Link - Download Client SDK

Community Features

WSO2

Engage with the Community

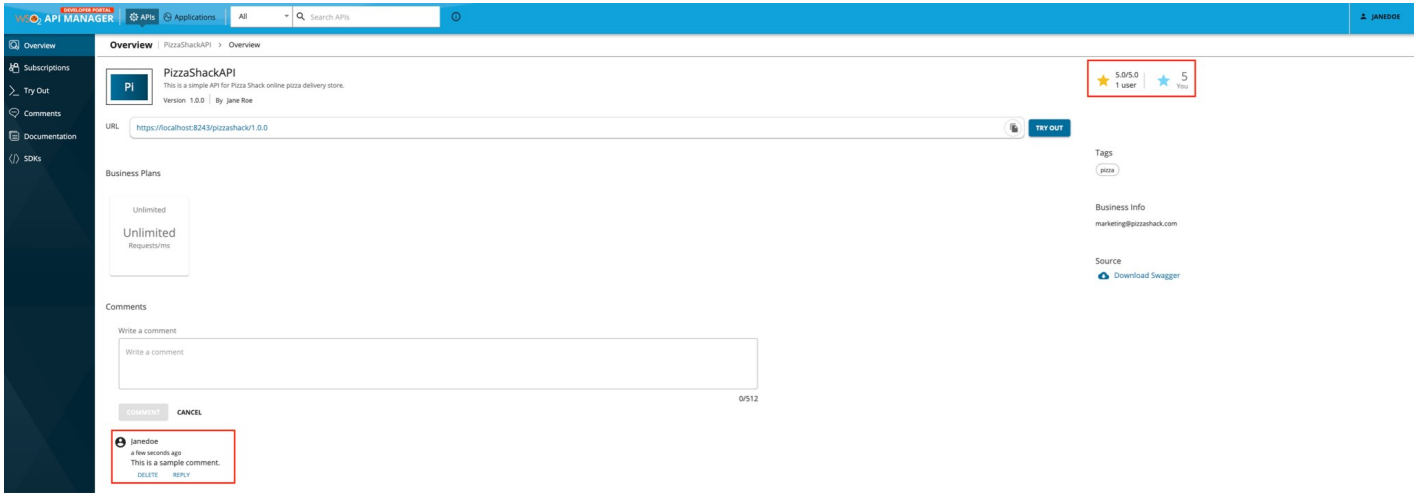
The Developer Portal provides several useful features to build and nurture an active community of users as well as a developer community, for your APIs.

- Rate and comment
- Search facility

Link - Using the Community Features

Rate and Comment

Give insights to potential API consumers on the quality and usefulness of an API



Link - Rate and Comment

Let’s try it out!

Subscribing to APIs

Invoking the API

WSO2